
Los Sabrossos

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1. Math

1.1. Sieve of Eratosthenes

```
1  const int N = 1e7;
2  int sieve[N];
3  void make(){
4      for(int i = 0; i < N; i++) sieve[i] = 1;
5      sieve[0] = sieve[1] = 0;
6      for (int i = 2; i < N; i++)
7          {
8              if(sieve[i]){
9                  for (int j = i*i; j < N; j+=i)
10                     {
11                         sieve[j] = 0;
12                     }
13             }
14         }
15 }
```

1.2. Euler Phi

```
1  ll phi(ll n) {
2      ll ans = n;
3      for (ll p = 2; p * p <= n; p++) {
4          if (n % p == 0) {
5              ans -= ans / p;
6              while (n % p == 0){
7                  n /= p;
8              }
9          }
10     }
11     if (n > 1){
12         ans -= ans / n;
13     }
14     return ans;
15 }
```

1.3. Binary Exponentiation

```
1  long long pote(long long a, long long b, long long mod){
2      long long ans = 1ll;
3      while(b){
4          if(b&1) ans = (ans * a)%mod;
5          a = (a * a)%mod;
6          b>>=1;
7      }
8      return ans;
9  }
10 // a^-1 = a^mod-2 = pote(a, mod-2);
```

1.4. Extended Euclides

```
1  //Encuentra
2  //valores para x e y
3  //A*x + B*y = gcd(A, B)
4  ll extgcd(ll a, ll b, ll &x, ll &y){
5      if(b == 0){
6          x = 1;
7          y = 0;
8          return a;
9      }
10
11     ll x1, y1;
12     ll g = extgcd(b, a % b, x1, y1);
13     x = y1;
14     y = x1 - (a / b) * y1;
15     return g;
16 }
17
18 void solve(){
19     ll A, B, C;
20     cin >> A >> B >> C;
21     ll x, y;
22     ll g = extgcd(A, B, x, y);
```

```
23     if(C % g != 0){
24         cout << "-1\n";
25         return;
26     }
27     ll k = -C / g;
28     x *= k;
29     y *= k;
30     cout << x << "□" << y << '\n';
31 }
```

1.5. Fibonacci Log(n)

```
1  long long int fibo(int n) {
2      long long int a, b, x, y, tmp;
3      a = x = tmp = 0;
4      b = y = 1;
5      while (n != 0)
6          if (n % 2 == 0) {
7              tmp = a * a + b * b;
8              b = b * a + a * b + b * b;
9              a = tmp;
10             n = n / 2;
11         } else {
12             tmp = a * x + b * y;
13             y = b * x + a * y + b * y;
14             x = tmp;
15             n = n - 1;
16         }
17         // x = fibo(n-1)
18         return y; // y = fibo(n)
19 }
```

1.6. Modular Inverse

```
1  const long long N = 1e6 + 10;
2  const int MOD = 1e9+7;
3  long long fact[N];
4  long long invfact[N];
5  void init(){
6      fact[0] = 1;
7      for(long long i = 1; i < N; i++)
8          fact[i] = fact[i-1] * i % MOD;
9
10     invfact[N-1] = pote(fact[N-1], MOD-2);
11
12     for(long long i = N-2; i >= 0; i--)
13         invfact[i] = invfact[i+1] * (i+1) % MOD;
14 }
15
16 long long nCk(long long n, long long k){
17     if(k < 0 || k > n) return 0;
18     return fact[n] * invfact[k] % MOD * invfact[n-k] % MOD;
19 }
```

1.7. Pollard Rho Rabin Miller

```
1  map<long long, long long> fact;
2
3  long long mcd(long long a, long long b) {
4      a = abs(a);
5      b = abs(b);
6      while (a != 0) {
7          long long r = b % a;
8          b = a;
9          a = r;
10     }
11     return b;
12 }
13 //Rabin Miller
14 bool primo(long long n) {
15     if (n < 2) return false;
16     if (n == 2) return true;
17     long long D = n - 1, S = 0;
18     while (D % 2 == 0) {
19         D /= 2;
20         S++;
21     }
```

```
22 static const long long STEPS = 1611;
23 for(long long pasos = 0; pasos < STEPS; pasos++){
24     const long long A = 1 + rand() % (n - 1);
25     long long M = pote(A, D, n);
26     if (M == 1 || M == (n - 1)) goto next;
27     for(long long k = 0; k < S-1; k++){
28         M = ((M) * M) % n;
29         if (M == (n - 1)) goto next;
30     }
31     return false;
32 next;;
33 }
34 return true;
35 }
36
37 //Rho de pollard
38 long long factor(long long n) {
39     static long long A, B;
40     A = 1 + rand() % (n - 1);
41     B = 1 + rand() % (n - 1);
42     #define fun(x) (((x) * (x + B)) % n + A)
43
44     long long x = 2, y = 2, d = 1;
45     while (d == 1 || d == -1) {
46         x = fun(x);
47         y = fun(fun(y));
48         d = mcd(x - y, n);
49     }
50     return abs(d);
51 }
52
53 void factorize(long long n){
54     assert(n > 0);
55     while (n > 1 && !primo(n)) {
56         long long fx;
57         do {
58             fx = factor(n);
59         } while (fx == n);
60
61         n /= fx;
62         factorize(fx);
63
64         for (auto &it : fact) {
65             while (n % it.first == 0) {
66                 n /= it.first;
67                 it.second++;
68             }
69         }
70     }
71     if (n > 1) fact[n]++;
72 }
```

2. Geometry

2.1. Convex Hull

```
1 // Devuelve sentido horario
2 vector<point> hull(vector<point> p)
3 {
4     int n = p.size();
5     vector<point> h;
6     sort(p.begin(), p.end());
7     for(int i = 0; i < n; i++){
8         while(h.size() >= 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
9             h.pop_back();
10        }
11        h.push_back(p[i]);
12    }
13    h.pop_back();
14    int k = h.size();
15    for(int i = n-1; i >= 0; i--){
16        {
17            while(h.size() >= k + 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
18                h.pop_back();
19            }
20            h.pb(p[i]);
21        }
22        h.pop_back();
23        return h;
24    }
```

2.2. Point

```

1 struct point
2 {
3     int x, y;
4     point() {}
5     point(int x, int y): x(x), y(y) {}
6     point operator -(point p) {return point(x - p.x, y - p.y);}
7     point operator +(point p) {return point(x + p.x, y + p.y);}
8     int sq() const {return x * x + y * y;}
9     double abs() const {return sqrt(sq());}
10    int operator ^(point p) {return x * p.y - y * p.x;}
11    int operator *(point p) {return x * p.x + y * p.y;}
12    point operator *(int a) {return point(x * a, y * a);}
13    bool operator <(const point& p) const {return x == p.x ? y < p.y : x < p.x;}
14    int left(point a, point b) {
15        return ((b - a) ^ (*this - a)); // 0, -, +
16    }
17 };
18 ostream& operator<<(ostream& os, const point& p) {
19     return os << "(" << p.x << ", " << p.y << ")\n";
20 }
21
22
23 void polarSort(vector<point>& v) {
24     sort(v.begin(), v.end(), [] (point a, point b) {
25         const point origin{0, 0};
26         bool ba = a < origin, bb = b < origin;
27         if (ba != bb) { return ba < bb; }
28         return (a^b) > 0;
29     });
30 }

```

2.3. Closest Pair of Points Lineal

```

1 struct pt {
2     ll x, y;
3     bool operator == (const pt & o) const {
4         return x == o.x and y == o.y;
5     }
6 };
7
8 struct CustomHashPoint {
9     size_t operator()(const pt & p) const {
10         static
11         const uint64_t C =
12             chrono::steady_clock::now().time_since_epoch().count();
13         uint64_t x = p.x;
14         uint64_t y = p.y;
15         x ^= x >> 30;
16         x *= 0xbf58476d1ce4e5b9;
17         x ^= x >> 27;
18         x *= 0x94d049bb133111eb;
19         x ^= x >> 31;
20         y ^= y >> 30;
21         y *= 0xbf58476d1ce4e5b9;
22         y ^= y >> 27;
23         y *= 0x94d049bb133111eb;
24         y ^= y >> 31;
25         return C ^ (x + (y << 1));
26     }
27 };
28
29 ll dist2(pt a, pt b) {
30     ll x = a.x - b.x;
31     ll y = a.y - b.y;
32     return x * x + y * y;
33 }
34
35 pair < int, int > closest_pair(vector < pt > p) {
36     int n = p.size();
37     unordered_map < pt, int, CustomHashPoint > mp;
38     // puntos repetidos
39     for (int i = 0; i < n; i++) {
40         if (mp.count(p[i]))
41             return {
42                 mp[p[i]],
43                 i
44             };
45         mp[p[i]] = i;
46     }
47     mt19937 rng(
48         chrono::steady_clock::now().time_since_epoch().count()

```

```

49     };
50     auto rnd = [&]() {
51         return uniform_int_distribution < int > (0, n - 1)(rng);
52     };
53     ll d2 = dist2(p[0], p[1]);
54     pair < int, int > ans = {
55         0,
56         1
57     };
58     auto upd = [&](int i, int j) {
59         ll nd = dist2(p[i], p[j]);
60         if (nd < d2) {
61             d2 = nd;
62             ans = {
63                 i,
64                 j
65             };
66         }
67     };
68     // conseguir una distancia inicial razonable
69     for (int it = 0; it < 20; it++) {
70         int i = rnd();
71         int j = rnd();
72         while (i == j) j = rnd();
73         upd(i, j);
74     }
75     ll d = sqrtl((long double) d2) + 1;
76     unordered_map < pt, vector < int >, CustomHashPoint > grid;
77     grid.reserve(2 * n);
78     for (int i = 0; i < n; i++) {
79         grid[
80             p[i].x / d,
81             p[i].y / d
82         ].push_back(i);
83     }
84     for (auto & [c, v]: grid) {
85         // misma celda
86         for (int i = 0; i < (int) v.size(); i++) {
87             for (int j = i + 1; j < (int) v.size(); j++) {
88                 upd(v[i], v[j]);
89             }
90         }
91         // celdas vecinas
92         for (int dx = -1; dx <= 1; dx++) {
93             for (int dy = -1; dy <= 1; dy++) {
94                 if (dx == 0 and dy == 0) continue;
95                 pt to = {
96                     c.x + dx,
97                     c.y + dy
98                 };
99                 auto it = grid.find(to);
100                 if (it == grid.end()) continue;
101                 for (int i: v) {
102                     for (int j: it->second) {
103                         upd(i, j);
104                     }
105                 }
106             }
107         }
108     }
109     return ans;
110 }

```

2.4. Closest Pair of Points Nlogn

```

1  struct pt {
2      ll x, y;
3      int id;
4  };
5
6  ll dist2(pt a, pt b) {
7      ll x = a.x - b.x;
8      ll y = a.y - b.y;
9      return x * x + y * y;
10 }
11
12 pair < int, int > closest_pair(vector < pt > p) {
13     int n = p.size();
14     sort(p.begin(), p.end(), [](pt a, pt b) {
15         if (a.x != b.x) return a.x < b.x;
16         return a.y < b.y;
17     });
18     set < pair < ll, int >> s; // {y, id}
19     ll d2 = dist2(p[0], p[1]);

```



```

20     pair < int, int > ans = {
21         p[0].id,
22         p[1].id
23     };
24     int l = 0;
25     for (int i = 0; i < n; i++) {
26         ll d = sqrtl((long double) d2) + 1;
27         // sacar puntos que ya est n demasiado lejos en x
28         while (l < i and p[i].x - p[l].x >= d) {
29             s.erase({
30                 p[l].y,
31                 p[l].id
32             });
33             l++;
34         }
35         // buscar puntos con y cerca
36         auto it = s.lower_bound({
37             p[i].y - d,
38             -1
39         });
40         while (it != s.end() and it -> first <= p[i].y + d) {
41             int j = it -> second;
42             ll nd = dist2(p[i], p[j]);
43             if (nd < d2) {
44                 d2 = nd;
45                 ans = {
46                     p[i].id,
47                     p[j].id
48                 };
49                 d = sqrtl((long double) d2) + 1;
50             }
51             it++;
52         }
53         s.insert({
54             p[i].y,
55             p[i].id
56         });
57     }
58     return ans;
59 }

```

3. Strings

3.1. KMP

```

1     vector<int> kmp(string s){
2         int n = s.size();
3         vector<int> pi(n);
4         for(int i = 1; i < n; i++){
5             int j = pi[i-1];
6             while(j > 0 and s[i] != s[j]){
7                 j = pi[j-1];
8             }
9             if(s[i] == s[j]){
10                j++;
11            }
12            pi[i] = j;
13        }
14        return pi;
15    }
16    //Pattern Matching
17    vector<int> ocurrencia(string texto, string patron) {
18        vector<int> P = kmp(patron);
19        int k = 0;
20        vector<int> M;
21        repl(i,0,texto.size()) {
22            while (k > 0 and patron[k] != texto[i]) k = P[k - 1];
23            if (patron[k] == texto[i]) k++;
24            if (k == patron.size()) {
25                M.pb(i - k + 1);
26                k = P[k - 1];
27            }
28        }
29        return M;
30    }
31
32    //Automata
33    void genAut(string &s) {
34        int n = s.size();
35
36        vector<vector<int>>> aut(n+1, vector<int>(26));

```

```

37     vector<int> pi = kmp(s);
38
39     for(int i = 0 ; i < n; i++){
40         aut[i][s[i]-'a'] = i+1;
41     }
42
43     for(int i = 0; i < n+1; i++){
44         for(int c = 0; c < 26; c++){
45             if (aut[i][c]) continue;
46             if (i < n and s[i] == 'a' + c) aut[i][c] = i+1;
47             else if (i > 0) aut[i][c] = aut[pi[i-1]][c];
48             else aut[i][c] = 0;
49         }
50     }
51 }

```

3.2. Z Function

```

1  vector<int> z_function(string s){
2      int n = s.size();
3      vector<int> z(n,0);
4      int l = 0;
5      int r = 0;
6      repl(i,l,n){
7          if(i < r){
8              z[i] = min(r - i, z[i - l]);
9          }
10         while(i + z[i] < n and s[z[i]] == s[i + z[i]]){
11             z[i]++;
12         }
13         if(i + z[i] > r){
14             l = i;
15             r = i + z[i];
16         }
17     }
18     return z;
19 }

```

3.3. Trie

```

1  const int E = 26;
2
3  struct AC {
4      int nodes;
5      vector<array<int, E>>go;
6      vector<bool>terminal;
7
8      AC(){
9          add_node();
10         nodes = 1;
11     }
12
13     void add_node(){
14         terminal.emplace_back();
15         go.emplace_back();
16     }
17
18     void insert(string &s){
19         int pos = 0;
20         for(char ch : s){
21             int c = ch - '0';
22             if(go[pos][c] == 0){
23                 go[pos][c] = nodes++;
24                 add_node();
25             }
26             pos = go[pos][c];
27         }
28         terminal[pos] = true;
29     }
30
31 };

```

3.4. Suffix Array

```

1  vector<int> suffix_array(string s) {
2      s += char(0);
3      const int n = s.size();

```

```

4     vector<int>a(n);
5     iota(a.begin(), a.end(), 0);
6     sort(a.begin(), a.end(), [&] (int i, int j){
7         return s[i] < s[j];
8     });
9     vector<int> mapping(n);
10    mapping[a[0]] = 0;
11    for(int i = 1; i < n; i++){
12        mapping[a[i]] = mapping[a[i-1]]+(s[a[i-1]] != s[a[i]]);
13    }
14    int len = 1;
15    vector<int>sbs(n);
16    vector<int>head(n);
17    vector<int>new_mapping(n);
18    while(len < n){
19        for(int i = 0; i<n; i++) sbs[i] = (a[i] + n - len) % n;
20        for(int i = n-1; i>=0; i--) head[mapping[a[i]]] = i;
21        for(int i = 0; i < n; i++){
22            int x = sbs[i];
23            a[head[mapping[x]]++] = x;
24        }
25        new_mapping[a[0]] = 0;
26        for(int i = 1; i < n; i++){
27            if(mapping[a[i-1]] != mapping[a[i]]){
28                new_mapping[a[i]] = new_mapping[a[i-1]]+1;
29            }
30            else{
31                int pre = mapping[(a[i-1] + len)%n];
32                int cur = mapping[(a[i]+len) % n];
33                new_mapping[a[i]] = new_mapping[a[i-1]] + (pre!=cur);
34            }
35        }
36        len <= 1;
37        swap(mapping, new_mapping);
38    }
39    return vector<int>(a.begin()+1, a.end()); //ignorar a[0] que es el centinela
40 }
41
42 vector<int> lcp_array(vector<int>&a, string s){
43     const int n = s.size();
44     vector<int> rank(n);
45     for(int i = 0; i < n; i++) rank[a[i]] = i;
46     int k = 0;
47     vector<int>lcp(n);
48     for(int i = 0; i < n; i++){
49         if(rank[i] == n-1){
50             k = 0;
51             continue;
52         }
53         int j = a[rank[i]+1];
54         while(i+k < n and j+k < n and s[i+k] == s[j+k]) k++;
55         lcp[rank[i]] = k;
56         if(k) k--;
57     }
58     lcp.pop_back();
59     return lcp;
60 }
61
62 void solve(){
63     //Numero de distintos substr en un string  $n*(n+1)/2$  - sumatoria del LCP
64     //verificar si un string es substring de otro con BS en SA
65     //encontrar el longest common substring en 2 strings (ESTA IMPLEMENTACION)
66     string s, t;
67     cin >> s >> t;
68     if(s.size() > t.size()){
69         swap(s, t);
70     }
71     int n = s.size();
72     int m = t.size();
73     string st = s+'#'+t;
74     vector<int>sa = suffix_array(st);
75     vector<int>lcp = lcp_array(sa, st);
76     // print(sa);
77     // print(lcp);
78     vector<int>tr(n+m);
79     repl(i,0,n+m){
80         if(sa[i] >=n and sa[i+1]>=n){
81             tr[i] = 0;
82             continue;
83         }
84         if(sa[i] < n and sa[i+1] < n){
85             tr[i] = 0;
86             continue;
87         }
88         int mn = min(sa[i], sa[i+1]);
89         int tam1 = mn + lcp[i];

```

```

90         // dbg(tam1);
91         tr[i] = lcp[i];
92         if(tam1 > n){
93             dbg(tam1);
94             tr[i] = lcp[i] - (tam1-n);
95         }
96     }
97     // print(tr);
98     int mx = *max_element(tr.begin(), tr.end());
99     repl(i,0,n+m){
100         // dbg(tr[i]);
101         if(tr[i] == mx){
102             string ans = st.substr(sa[i], mx);
103             cout << ans << '\n';
104             return;
105         }
106     }
107 }
108 //Sabrossus
109

```

3.5. Aho Corasick

```

1 //AhoCorasick para saber si un string s existe en un texto T
2 const int E = 26;
3 const int N = 1e6+10;
4 int cur;
5 int nodoTerminal[N]; //se guarda el nodo terminal del string i
6 set<int>st; //se guarda que nodos son terminales que aparecen en el texto T
7
8 struct AC {
9     int nodes;
10    vector<int>suf, super;
11    vector<array<int, E>>go;
12    vector<bool>terminal;
13    AC(){
14        add_node();
15        nodes = 1;
16    }
17    void add_node(){
18        terminal.emplace_back();
19        go.emplace_back();
20        super.emplace_back();
21        suf.emplace_back();
22    }
23
24    void insert(string &s){
25        int pos = 0;
26        for(char ch : s){
27            int c = ch - 'a';
28            if(go[pos][c] == 0){
29                go[pos][c] = nodes++;
30                add_node();
31            }
32            pos = go[pos][c];
33        }
34        terminal[pos] = true;
35        nodoTerminal[cur] = pos;
36    }
37
38    void build(){
39        queue<int>Q;
40        Q.emplace(0);
41        while(!Q.empty()){
42            int u = Q.front();
43            Q.pop();
44            super[u] = (suf[u] == 0 or terminal[suf[u]] ? suf[u] : super[suf[u]]);
45            for(int c = 0; c < E; c++){
46                if(go[u][c]){
47                    int v = go[u][c];
48                    Q.emplace(v);
49                    suf[v] = u == 0 ? 0 : go[suf[u]][c];
50                }
51                else{
52                    go[u][c] = u == 0 ? 0 : go[suf[u]][c];
53                }
54            }
55        }
56    }
57 };
58
59 void mark(int u, AC &ac){
60     if(u == 0) return;
61     mark(ac.super[u], ac);

```

```

62     if(ac.terminal[u]){
63         st.insert(u);
64         ac.terminal[u] = false;
65     }
66     ac.super[u] = 0;
67 }
68
69 void solve(){
70     int n;
71     cin >> n;
72     AC ac;
73     repl(i,0,n){
74         string s;
75         cin >> s;
76         cur = i;
77         ac.insert(s);
78     }
79     string t; //texto grande
80     cin >> t;
81     ac.build();
82     int p = 0;
83     for(auto x : t){
84         int c = x - 'a';
85         p = ac.go[p][c];
86         mark(p, ac);
87     }
88     repl(i,0,n){
89         if(st.count(nodoTerminal[i])){
90             cout << "YES\n";
91         }
92         else{
93             cout << "NO\n";
94         }
95     }
96 }
97
98 //-----
99
100 //AhoCorasick para Frecuencias
101 const int E = 26;
102 const int N = 1e6+10;
103 int nodoTerminal[N];
104 int cur;
105 struct AC {
106     int nodes;
107     vector<int> suf, freq, by_level;
108     vector<bool> terminal;
109     vector<array<int,E>> go;
110
111     AC(){
112         add_node();
113         nodes = 1;
114     }
115     void add_node(){
116         terminal.emplace_back();
117         suf.emplace_back();
118         go.emplace_back();
119         freq.emplace_back();
120     }
121
122     void insert(string &s){
123         int pos = 0;
124         for(char ch:s){
125             int c = ch - 'a';
126             if(go[pos][c] == 0){
127                 go[pos][c] = nodes++;
128                 add_node();
129             }
130             pos = go[pos][c];
131         }
132         terminal[pos] = true;
133         nodoTerminal[cur] = pos;
134     }
135
136     void build(){
137         queue<int> Q;
138         Q.emplace(0);
139         while(!Q.empty()){
140             int u = Q.front();
141             Q.pop();
142             by_level.emplace_back(u);
143             for(int c = 0; c < E; c++){
144                 if(go[u][c]){
145                     int v = go[u][c];
146                     Q.emplace(v);
147                     suf[v] = u == 0 ? 0 : go[suf[u]][c];

```

```

148         }
149         else{
150             go[u][c] = u == 0 ? 0 : go[suf[u]][c];
151         }
152     }
153 }
154 };
155 };
156
157 void solve(){
158     int n;
159     cin >> n;
160     AC ac;
161     repl(i,0,n){
162         string s;
163         cin >> s;
164         cur = i;
165         ac.insert(s);
166     }
167     string t;
168     cin >> t;
169     ac.build();
170     int p = 0;
171     for(char ch:t){
172         int c = ch-'a';
173         p = ac.go[p][c];
174         ac.freq[p]++;
175     }
176     for(int i = (int)ac.by_level.size()-1; i > 0; i--){
177         int x = ac.by_level[i];
178         ac.freq[ac.suf[x]] += ac.freq[x];
179     }
180     repl(i,0,n){
181         int termi = nodoTerminal[i];
182         cout << ac.freq[termi] << '\n';
183     }
184 }

```

3.6. Hashing

```

1 struct Hash {
2     long long P=1777771,MOD[2],PI[2];
3     vector<long long> h[2],pi[2];
4
5     // SOLO PARA ROLLING / JOIN:
6     vector<long long> pw[2];
7
8     Hash(string& s){
9         MOD[0]=999727999;
10        MOD[1]=107077777;
11        PI[0]=325255434;
12        PI[1]=10018302;
13
14        for(int k = 0; k < 2; k++) {
15            h[k].resize(s.size()+1);
16            pi[k].resize(s.size()+1);
17
18            // SOLO PARA ROLLING / JOIN:
19            pw[k].resize(s.size()+1);
20        }
21
22        for(int k = 0; k < 2; k++){
23            h[k][0] = 0;
24            pi[k][0] = 1;
25
26            // SOLO PARA ROLLING / JOIN:
27            pw[k][0] = 1;
28
29            long long p = 1;
30
31            for(int i = 1; i < s.size()+1; i++){
32                h[k][i] = (h[k][i-1]+p*s[i-1])%MOD[k];
33                pi[k][i] = (1LL*pi[k][i-1]*PI[k])%MOD[k];
34
35                // SOLO PARA ROLLING / JOIN:
36                pw[k][i] = (1LL*pw[k][i-1]*P)%MOD[k];
37
38                p = (p*P)%MOD[k];
39            }
40        }
41    }
42
43    long long get(int s, int e){
44        long long h0 = (h[0][e]-h[0][s]+MOD[0])%MOD[0];

```

```
45         h0 = (1LL*h0*pi[0][s])%MOD[0];
46
47         long long h1 = (h[1][e]-h[1][s]+MOD[1])%MOD[1];
48         h1 = (1LL*h1*pi[1][s])%MOD[1];
49
50         return (h0<<32)|h1;
51     }
52
53     // SOLO PARA ROLLING / JOIN:
54     long long join(long long a, long long b, int lenA){
55         long long a0 = a>>32;
56         long long a1 = a&0xffffffffLL;
57
58         long long b0 = b>>32;
59         long long b1 = b&0xffffffffLL;
60
61         a0 = (a0 + b0*pw[0][lenA])%MOD[0];
62         a1 = (a1 + b1*pw[1][lenA])%MOD[1];
63
64         return (a0<<32)|a1;
65     }
66 };
```

3.7. Manacher

```
1  const int N = 1e5 + 10;
2  int odd[N]; //odd[i] = max odd palindrome centered on i
3  int even[N]; //even[i] = max even palindrome centered on i
4  void manacher(string & s) {
5      int l = 0, r = -1, n = s.size();
6      for (int i = 0; i < n; i++) {
7          int k = i > r ? 1 : min(odd[l + r - i], r - i);
8          while (i + k < n and i - k >= 0 and s[i + k] == s[i - k]) k++;
9          odd[i] = k--;
10         if (i + k > r) l = i - k, r = i + k;
11     }
12     l = 0;
13     r = -1;
14     for (int i = 0; i < n; i++) {
15         int k = i > r ? 0 : min(even[l + r - i + 1], r - i + 1);
16         k++;
17         while (i + k <= n and i - k >= 0 and s[i + k - 1] == s[i - k]) k++;
18         even[i] = --k;
19         if (i + k - 1 > r) l = i - k, r = i + k - 1;
20     }
21 }
```

4. Bit Manipulation

4.1. Binary Representation

```
1  string tobit(long long x) {
2      string a(64, '0');
3      for (int i = 63; i >= 0; --i){
4          if (x & 1) a[i] = '1';
5          x >>= 1;
6      }
7      return a;
8  }
```

4.2. Bit Mask

```
1  for(int mask = 0; mask < (1 << n); mask++) { //(1 << n) = 0(2^n)
2      for(int i = 0; i < n; i++) { // 0(n)
3          //verificamos si el i-esimo bit esta encendido
4          if(mask & (1 << i)) {
5              cout << v[i] << " ";
6          }
7      }
8  }
```

4.3. Bits Prendidos En Un Numero X En Una Posicion Pos

```
1 //retorna el numero de bits prendidos
2 //en una posicion pos desde 0 hasta el numero num
3 //tambien se puede usar en rangos asi:
4 //prendidos(R, 1) - prendidos(L-1, 1);
5 ll prendidos(ll num, ll pos){
6     if(num <= 0){
7         return 0;
8     }
9     ll pote = (1ll << pos);
10    ll blocks = (num + 1) / pote;
11    ll ones = blocks / 2;
12    ones *= pote;
13    if(blocks & 1){
14        ones += (num + 1) % pote;
15    }
16    return ones;
17 }
18 //Sabrosson
```

5. Data Structures

5.1. Segment Tree

```
1 const int N = 1e5;
2
3 struct info {
4     int num;
5 };
6
7 info ST[4 * N];
8
9 info ope(info u, info v) {
10     return {u.num + v.num};
11 }
12
13 void build(const vector<int> &v, int in = 1, int L = 0, int R = n - 1) {
14     if (L == R) {
15         ST[in] = {v[L]}; // Asignamos el valor desde el vector
16     } else {
17         int mid = (L + R) >> 1;
18         build(v, 2 * in, L, mid);
19         build(v, (2 * in) + 1, mid + 1, R);
20         ST[in] = ope(ST[2 * in], ST[(2 * in) + 1]);
21     }
22 }
23
24 void update(int pos, int val, int in = 1, int L = 0, int R = n - 1) {
25     if (L == R) {
26         ST[in].num = val;
27     } else {
28         int mid = (L + R) >> 1;
29         if (pos <= mid) {
30             update(pos, val, 2 * in, L, mid);
31         } else {
32             update(pos, val, (2 * in) + 1, mid + 1, R);
33         }
34         ST[in] = ope(ST[2 * in], ST[(2 * in) + 1]);
35     }
36 }
37
38 info get(int l, int r, int in = 1, int L = 0, int R = n - 1) {
39     if (r < L or R < l) return {0}; // Elemento neutro
40     if (l <= L and R <= r) return ST[in];
41
42     int mid = (L + R) >> 1;
43     info left = get(l, r, 2 * in, L, mid);
44     info right = get(l, r, (2 * in) + 1, mid + 1, R);
45     return ope(left, right);
46 }
47
48 // Llamadas simples y limpias:
49 // build(v); // Construye con el vector
50 // update(2, 10); // Cambia el valor en pos 2 a 10
51 // info ans = get(1, 3); // Obtiene la respuesta en el rango [1, 3]
52 // v = {1 2 3 4 5}
53 // get(1, 0, n-1, 1, 3) = 2 + 3 + 4 = 9
```


5.2. Segment Tree Lazy Propagation

```

1  const int N = 1e5;
2
3  struct info {
4      long long num;
5  };
6
7  info ST[4 * N];
8  long long lazy[4 * N];
9
10 info ope(info u, info v) {
11     return {u.num + v.num};
12 }
13
14 void build(const vector<int> &v, int in = 1, int L = 0, int R = n - 1) {
15     if (L == R) {
16         ST[in] = {v[L]};
17     } else {
18         int mid = (L + R) >> 1;
19
20         build(v, 2 * in, L, mid);
21         build(v, 2 * in + 1, mid + 1, R);
22
23         ST[in] = ope(ST[2 * in], ST[2 * in + 1]);
24     }
25 }
26
27 void push(int in, int L, int R) {
28     if (lazy[in] == 0)
29         return;
30
31     ST[in].num += lazy[in] * (R - L + 1);
32
33     if (L != R) {
34         lazy[2 * in] += lazy[in];
35         lazy[2 * in + 1] += lazy[in];
36     }
37
38     lazy[in] = 0;
39 }
40
41 void update(int l, int r, long long val,
42             int in = 1, int L = 0, int R = n - 1) {
43
44     push(in, L, R);
45
46     if (r < L or R < l)
47         return;
48
49     if (l <= L and R <= r) {
50         lazy[in] += val;
51         push(in, L, R);
52         return;
53     }
54
55     int mid = (L + R) >> 1;
56
57     update(l, r, val, 2 * in, L, mid);
58     update(l, r, val, 2 * in + 1, mid + 1, R);
59
60     ST[in] = ope(ST[2 * in], ST[2 * in + 1]);
61 }
62
63 info get(int l, int r,
64          int in = 1, int L = 0, int R = n - 1) {
65
66     push(in, L, R);
67
68     if (r < L or R < l)
69         return {0};
70
71     if (l <= L and R <= r)
72         return ST[in];
73
74     int mid = (L + R) >> 1;
75
76     info left = get(l, r, 2 * in, L, mid);
77     info right = get(l, r, 2 * in + 1, mid + 1, R);
78
79     return ope(left, right);
80 }

```

5.3. Bit 1D

```

1  struct BIT{
2      int n;
3      vector<ll> bit;
4      BIT(int _n){
5          n = _n;
6          bit.assign(n + 1, 0);
7      }
8      void add(int x, ll val){
9          for (int i = x; i <= n; i += (i&(-i))){
10             bit[i] += val;
11         }
12     }
13     ll sum(int x){
14         ll ans = 0ll;
15         for (int i = x; i > 0; i -= (i&(-i))){
16             ans += bit[i];
17         }
18         return ans;
19     }
20     ll query(int l, int r){
21         return sum(r) - sum(l - 1);
22     }
23 }
24 };

```

5.4. Bit 2D

```

1  struct BIT{
2      int n, m;
3      vector<vector<ll>> bit;
4
5      BIT(int _n){
6          n = _n;
7          m = _n;
8          bit.assign(n+1, vector<ll>(n+1, 0));
9      }
10     void add(int x, int y, ll val){
11         for (int i = x; i <= n; i += (i&(-i))){
12             for (int j = y; j <= m; j += (j&(-j))){
13                 bit[i][j] += val;
14             }
15         }
16     }
17     ll sum(int x, int y){
18         ll ans = 0ll;
19         for (int i = x; i > 0; i -= (i&(-i))){
20             for (int j = y; j > 0; j -= (j&(-j))){
21                 ans += bit[i][j];
22             }
23         }
24         return ans;
25     }
26     ll query(int x1, int y1, int x2, int y2){
27         return sum(x2, y2) - sum(x2, y1 - 1) - sum(x1 - 1, y2) + sum(x1 - 1, y1 - 1);
28     }
29 };

```

5.5. Bit 3D

```

1  struct BIT {
2      int n;
3      int m;
4      int p;
5      vector < vector < vector < ll >>> bit;
6      BIT(int _n) {
7          n = _n + 10;
8          m = _n + 10;
9          p = _n + 10;
10         bit.assign(n, vector < vector < ll >> (n, vector < ll > (n, 0)));
11     }
12     void add(int x, int y, int z, int val) {
13         for (int i = x; i <= n; i += (i & (-i))) {
14             for (int j = y; j <= m; j += (j & (-j))) {
15                 for (int k = z; k <= p; k += (k & (-k))) {
16                     bit[i][j][k] += val;
17                 }
18             }
19         }
20     }

```

```

20     }
21     ll sum(int x, int y, int z) {
22         ll ans = 0;
23         for (int i = x; i > 0; i -= (i & (-i))) {
24             for (int j = y; j > 0; j -= (j & (-j))) {
25                 for (int k = z; k > 0; k -= (k & (-k))) {
26                     ans += bit[i][j][k];
27                 }
28             }
29         }
30         return ans;
31     }
32     ll query(int x1, int y1, int z1, int x2, int y2, int z2) {
33         return sum(x2, y2, z2) - sum(x1 - 1, y2, z2) - sum(x2, y2, z1 - 1) - sum(x2, y1 - 1, z2)
34             + sum(x1 - 1, y1 - 1, z2) + sum(x2, y1 - 1, z1 - 1) + sum(x1 - 1, y2, z1 - 1) -
35             sum(x1 - 1, y1 - 1, z1 - 1);
36     }
37 };

```

5.6. Disjoint Set Union

```

1  struct DSU{
2      int num;
3      vector<int> parent;
4      vector<int> sizes;
5      DSU(){};
6      DSU(int n){
7          init(n);
8      }
9      void init(int n){
10         num = n;
11         parent.resize(n+1);
12         sizes.resize(n+1);
13         //si tienes index 0 inicializar desde 0
14         for(int i = 1; i <= n ; i++){
15             parent[i] = i;
16             sizes[i] = 1;
17         }
18     }
19     int find(int a){
20         if(a == parent[a]) return a;
21         return parent[a] = find(parent[a]);
22     }
23     void join(int a, int b){
24         int x = find(a);
25         int y = find(b);
26         if(x == y) return;
27         num--;
28         if(sizes[x] < sizes[y]){
29             parent[x] = y;
30             sizes[y] += sizes[x];
31         }
32         else{
33             parent[y] = x;
34             sizes[x] += sizes[y];
35         }
36     }
37 };
38 //Sabrosssus

```

5.7. Implicit Treap

```

1  // Mantiene las posiciones
2  struct Node {
3      int val , pri , sz ;
4      Node *l , *r;
5      Node ( int v ) {
6          val = v;
7          pri = rand () ;
8          sz = 1;
9          l = r = nullptr ;
10     }
11 };
12 using pNode = Node *;
13 int sz ( pNode t ) {
14     return t ? t -> sz : 0;
15 }
16 void upd (pNode t) {
17     if (t) t -> sz = 1 + sz (t -> l) + sz (t -> r);
18 }

```

```

19 void split ( pnode t , pnode &l , pnode &r , int k ) {
20     if (! t) {
21         l = r = nullptr ;
22         return ;
23     }
24     int leftSize = sz ( t -> l );
25     if ( leftSize >= k ) {
26         split ( t -> l , l , t -> l , k ) ;
27         r = t ;
28     }
29     else {
30         split ( t -> r , t -> r , r , k - leftSize - 1 ) ;
31         l = t ;
32     }
33     upd ( t );
34 }
35 pnode merge ( pnode l , pnode r ) {
36     if (! l || ! r ) return l ? l : r ;
37
38     if ( l -> pri > r -> pri ) {
39         l -> r = merge ( l -> r , r );
40         upd ( l );
41         return l ;
42     }
43     else {
44         r -> l = merge ( l , r -> l );
45         upd ( r );
46         return r ;
47     }
48 }
49 void insert ( pnode & root , int pos , int val ) {
50     pnode a , b ;
51     split ( root , a , b , pos );
52     root = merge ( merge ( a , new Node ( val ) ) , b ) ;
53 }
54 void erase ( pnode & root , int pos ) {
55     pnode a , b , c ;
56     split ( root , a , b , pos );
57     split ( b , b , c , 1 );
58     root = merge ( a , c ) ;
59 }
60 void print(pnode t) {
61     if (!t) return;
62
63     print(t->l);
64     cout << t->val << " ";
65     print(t->r);
66 }
67 int get(pnode t, int k) {
68     if (!t) return -1;
69
70     int leftSize = sz(t->l);
71
72     if (k < leftSize)
73         return get(t->l, k);
74
75     if (k == leftSize)
76         return t->val;
77
78     return get(t->r, k - leftSize - 1);
79 }

```

5.8. Ordered Set

```

1  #include <bits/stdc++.h>
2  #include <ext/pb_ds/assoc_container.hpp>
3  #include <ext/pb_ds/tree_policy.hpp>
4
5  using namespace std;
6  using namespace __gnu_pbds;
7
8  typedef tree<int, null_type, less<int>, rb_tree_tag, tree_order_statistics_node_update>
9      ordered_set;
10
11 void solve() {
12     ordered_set st;
13
14     st.insert(10);
15     st.insert(30);
16     st.insert(20);
17
18     // 1. Elemento en la posición K (0-indexed, de menor a mayor)
19     // find_by_order devuelve un iterador (puntero). Usamos '*' para el valor.
20     auto it = st.find_by_order(1);

```

```

20     if (it != st.end()) {
21         int elemento = *it; // Devuelve 20 de forma segura
22     }
23
24     // 2. Cuantos elementos son estrictamente menores que X
25     int menores_que_25 = st.order_of_key(25); // Devuelve 2 (el 10 y el 20)
26 }
27 // Sabrossus

```

5.9. Sparse Table

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int v[N]; //values
4  int st[N][LOG]; //Sparse Table
5  int lg[N]; // log values
6  int n;
7  void build(){
8      for(int i = 0; i < n; i++) st[i][0] = v[i];
9
10     for(int i = 1; i < LOG; i++){
11         for(int j = 0; j < n - (1<<i) + 1; j++){
12             st[j][i] = min(st[j][i-1], st[j+(1<<(i-1))][i-1]);
13         }
14     }
15     lg[1] = 0;
16     for(int i = 2; i <= n; i++) lg[i] = lg[i/2] + 1;
17 }
18 int query(int L, int R){
19     int k = lg[R-L+1];
20     return min(st[L][k], st[R-(1<<k)+1][k]);
21 }
22 // v = [5, 2, 4, 7, 1, 3]
23 // query(1, 4)      [2, 4, 7, 1]

```

5.10. Treap

```

1  // Ordena los elementos por valor
2  struct Node {
3      int key, pri, sz;
4      Node *l, *r;
5      Node(int k) {
6          key = k;
7          pri = rand();
8          sz = 1;
9          l = r = nullptr;
10     }
11 };
12
13 using pnode = Node*;
14
15 int sz(pnode t) {
16     return t ? t->sz : 0;
17 }
18
19 void upd(pnode t) {
20     if (t) t->sz = 1 + sz(t->l) + sz(t->r);
21 }
22
23 void split(pnode t, pnode &l, pnode &r, int k) {
24     if (!t) {
25         l = r = nullptr;
26         return;
27     }
28
29     int leftSize = sz(t->l);
30
31     if (leftSize >= k) {
32         split(t->l, l, t->l, k);
33         r = t;
34     } else {
35         split(t->r, t->r, r, k - leftSize - 1);
36         l = t;
37     }
38
39     upd(t);
40 }
41
42 pnode merge(pnode l, pnode r) {
43     if (!l || !r) return l ? l : r;
44

```

```
45     if (l->pri > r->pri) {
46         l->r = merge(l->r, r);
47         upd(l);
48         return l;
49     } else {
50         r->l = merge(l, r->l);
51         upd(r);
52         return r;
53     }
54 }
55
56 void insert(pnode &root, int pos, int val) {
57     pnode a, b;
58     split(root, a, b, pos);
59     root = merge(merge(a, new Node(val)), b);
60 }
61
62 void erase(pnode &root, int pos) {
63     pnode a, b, c;
64     split(root, a, b, pos);
65     split(b, b, c, 1);
66     root = merge(a, c);
67 }
68
69 int get(pnode t, int k) {
70     if (!t) return -1;
71
72     int leftSize = sz(t->l);
73
74     if (k < leftSize)
75         return get(t->l, k);
76
77     if (k == leftSize)
78         return t->key;
79
80     return get(t->r, k - leftSize - 1);
81 }
82
83 void print(pnode t) {
84     if (!t) return;
85
86     print(t->l);
87     cout << t->key << " ";
88     print(t->r);
89 }
```

6. Dynamic Programing

6.1. Bounded Knapsack

```
1  // bounded_knapsack para DPs del estilo :
2  // dp[j - cw] + cv , w-> pesos-> valor y quieres maximizar el valor
3  int n, we;
4  cin >> n >> we;
5  vector<int> v, m, w, dp;
6  v.resize(n);
7  m.resize(n);
8  w.resize(n);
9  dp.assign(we+1, 0);
10 int sum = 0;
11 for(int i = 0 ; i < n ; i++){
12     int a, b, c;
13     cin >> a >> b >> c;
14     v[i] = a;
15     w[i] = b;
16     m[i] = c;
17     vector<int> old = dp;
18     for(int r = 0; r < w[i] and r <= we ; r++){
19         deque<pair<int,int>> dq;
20         for(int j = r; j <= we ; j += w[i]){
21             int k = (j - r)/w[i];
22             while(!dq.empty() and dq.front().first < k - m[i]){
23                 dq.pop_front();
24             }
25             int cur = old[j] - k * v[i];
26             while(!dq.empty() && dq.back().second <= cur){
27                 dq.pop_back();
28             }
29             dq.push_back({k, cur});
30             dp[j] = dq.front().second + k * v[i];
31         }
32     }
```

```
32     }
33 }
34 cout << dp[we];
```

6.2. Kadane 2D

```
1  int n;
2  cin >> n;
3  vector < vector < int >> v(n);
4  repl(i, 0, n) {
5      repl(j, 0, n) {
6          int x;
7          cin >> x;
8          v[i].push_back(x);
9      }
10 }
11 int maxi = INT_MIN;
12 repl(ini, 0, n) {
13     vector < int > pref(n, 0);
14     repl(i, ini, n) {
15         repl(j, 0, n) {
16             pref[j] += v[i][j];
17         }
18         int cur = 0;
19         repl(j, 0, n) {
20             cur = max(pref[j], pref[j] + cur);
21             maxi = max(maxi, cur);
22         }
23     }
24 }
```

6.3. Kadane 3D

```
1  int n;
2  cin >> n;
3
4  repl(i, 0, n) {
5      repl(j, 0, n) {
6          repl(k, 0, n) {
7              cin >> v[i][j][k];
8          }
9      }
10 }
11 int maxi = INT_MIN;
12 repl(k_ini, 0, n) {
13     vector < vector < int >> pref2(n, vector < int > (n, 0));
14     repl(k, k_ini, n) {
15         repl(i, 0, n) {
16             repl(j, 0, n) {
17                 pref2[i][j] += v[i][j][k];
18             }
19         }
20         repl(ini, 0, n) {
21             vector < int > pref(n, 0);
22             repl(i, ini, n) {
23                 repl(j, 0, n) {
24                     pref[j] += pref2[i][j];
25                 }
26                 int cur = 0;
27                 repl(u, 0, n) {
28                     cur = max(pref[u], cur + pref[u]);
29                     maxi = max(maxi, cur);
30                 }
31             }
32         }
33     }
34 }
35 }
36
37 cout << maxi << "\n";
```

6.4. SOS DP

```
1  void sos_dp(vector<ll>& dp, int K){
2      // k es el tamaño del arreglo :v
3      for(int bit = 0; bit < K; bit++){
4          for(int mask = 0; mask < (1 << K); mask++){
```

```
5         if(mask & (1 << bit)){
6             dp[mask] += dp[mask ^ (1 << bit)];
7             //      si quieres que sea un sos dp de maximos
8             //      dp[mask] = max(dp[mask], dp[mask ^ (1 << bit)]);
9         }
10    }
11 }
12 }
```

6.5. Fastfib

```
1  #include <bits/stdc++.h>
2  #define ll long long
3
4  using namespace std;
5
6  const ll MOD = 1e9 + 7;
7
8  struct Matrix {
9      ll a[2][2];
10     Matrix() {
11         memset(a, 0, sizeof(a));
12     }
13 };
14
15 Matrix multiply(Matrix A, Matrix B) {
16     Matrix C;
17     for (int i = 0; i < 2; i++) {
18         for (int j = 0; j < 2; j++) {
19             for (int k = 0; k < 2; k++) {
20                 C.a[i][j] = (C.a[i][j] + A.a[i][k] * B.a[k][j]) % MOD;
21             }
22         }
23     }
24     return C;
25 }
26
27 Matrix power(Matrix A, ll n) {
28     Matrix R;
29     R.a[0][0] = 1;
30     R.a[1][1] = 1;
31     while (n) {
32         if (n & 1)
33             R = multiply(R, A);
34         A = multiply(A, A);
35         n >>= 1;
36     }
37     return R;
38 }
39
40 ll fibonacci(ll n) {
41     if (n == 0) return 0;
42     Matrix A;
43     A.a[0][0] = 1;
44     A.a[0][1] = 1;
45     A.a[1][0] = 1;
46     A.a[1][1] = 0;
47     Matrix R = power(A, n - 1);
48     return R.a[0][0];
49 }
50
51 int main() {
52     ll n;
53     cin >> n;
54     cout << fibonacci(n) << '\n';
55 }
```

7. Flow

7.1. Dinic

```
1  struct Dinic
2  {
3      const int INF = 1e9;
4
5      struct flowEdge
6      {
7          int to, rev, f, cap;
```



```

8   };
9
10  vector<vector<flowEdge>> G;
11  vector<int> work, lvl, Q;
12
13  int S, T, nodes;
14
15  Dinic(){
16  Dinic(int n, int s, int t){
17      S = s;
18      T = t;
19      nodes = n;
20      G.clear();
21      G.resize(n);
22      Q.resize(n);
23  }
24  void addEdge(int st, int en, int cap){
25      flowEdge A = {en, (int)G[en].size(), 0, cap};
26      flowEdge B = {st, (int)G[st].size(), 0, 0};
27      G[st].pb(A);
28      G[en].pb(B);
29  }
30  int dfs(int v, int f){
31      if(v == T or f == 0) return f;
32      for(int &i = work[v]; i < G[v].size(); i++){
33          flowEdge &e = G[v][i];
34          int u = e.to;
35          if(e.cap <= e.f or lvl[u] != lvl[v] + 1) continue;
36          int df = dfs(u, min(f, e.cap - e.f));
37          if(df){
38              e.f += df;
39              G[u][e.rev].f -= df;
40              return df;
41          }
42      }
43      return 0;
44  }
45  bool bfs(){
46      int qt = 0;
47      Q[qt++] = S;
48      lvl.assign(nodes, -1);
49      lvl[S] = 0;
50      for(int i = 0; i < qt; i++){
51          int u = Q[i];
52          for(flowEdge &e : G[u]){
53              int v = e.to;
54              if(e.cap <= e.f or lvl[v] != -1) continue;
55              lvl[v] = lvl[u] + 1;
56              Q[qt++] = v;
57          }
58      }
59      return lvl[T] != -1;
60  }
61  int maxFlow(){
62      int flow = 0;
63      while(bfs()){
64          work.assign(nodes, 0);
65          while(true){
66              int df = dfs(S, INF);
67              if(df == 0) break;
68              flow += df;
69          }
70      }
71      return flow;
72  }
73
74  };

```

8. Graph Algorithms

8.1. Dijkstra

```

1   const int N = 1e5;
2
3   int vis[N];
4   vector<pair<int,int>> G[N];
5
6   void dijkstra(int x){
7       priority_queue<pair<int,int>, vector<pair<int,int>>, greater<pair<int,int>>> pq;
8       pq.push({0,x});
9       vis[x] = 0;

```

```

10 while(!pq.empty()){
11     auto y = pq.top();
12     pq.pop();
13     int u = y.second;
14     int cost = y.first;
15     for(auto v : G[u]){
16         int cur = cost + v.second;
17         if(cur < vis[v.first]){
18             vis[v.first] = cur;
19             pq.push({cur, v.first});
20         }
21     }
22 }
23 }

```

8.2. Kruskal MST

```

1  const int N = 1e5;
2  struct DSU{
3      int num;
4      vector<int> parent;
5      vector<int> sizes;
6      DSU(){};
7      DSU(int n){
8          init(n);
9      }
10     void init(int n){
11         num = n;
12         parent.resize(n+1);
13         sizes.resize(n+1);
14         for(int i = 1; i <= n ; i++){
15             parent[i] = i;
16             sizes[i] = 1;
17         }
18     }
19     int find(int a){
20         if(a == parent[a]) return a;
21         return parent[a] = find(parent[a]);
22     }
23     void join(int a, int b){
24         int x = find(a);
25         int y = find(b);
26         if(x == y) return;
27         num--;
28         if(sizes[x] < sizes[y]){
29             parent[x] = y;
30             sizes[y] += sizes[x];
31         }
32         else{
33             parent[y] = x;
34             sizes[x] += sizes[y];
35         }
36     }
37 };
38 int n;
39 vector<pair<int, pair<int, int>>> List;
40 int Kruskal_MST(){
41     sort(List.begin(), List.end());
42     DSU dsu(n);
43     int mst = 0, t = 1;
44     for(int i = 0; i < List.size(); i++){
45         int w = List[i].first; //weight
46         int a = List[i].second.first; //at
47         int b = List[i].second.second; //to
48         if(dsu.find(a) != dsu.find(b)){
49             dsu.join(a, b);
50             mst += w;
51             t++;
52         }
53         if(t == n) break;
54     }
55     return mst;
56 }

```

8.3. Topological Sort Kahn Algorithm

```

1  const int N = 1e5;
2
3  int indegree[N];
4  vector<int> G[N];
5  int n;

```

```

6  vector<int> Kahn(){
7      queue<int> q;
8      vector<int> ans;
9      for(int i = 0; i < n; i++){
10         if(indegree[i] == 0) q.push(i);
11     }
12     while (!q.empty()){
13         int v = q.front();
14         q.pop();
15         ans.push_back(v);
16         for(int u : G[v]){
17             indegree[u]--;
18             if(indegree[u] == 0){
19                 q.push(u);
20             }
21         }
22     }
23     //if(ans.size() != n) grafo no DAG
24     return ans;
25 }

```

9. Tree Algorithms

9.1. Euler Tour

```

1  const int N = 1e5 + 10;
2  vector<int> path;
3  vector<int> G[N];
4  void EulerTour(int node, int parent){
5      path.pb(node);
6      for(int u : G[node]){
7          if(u != parent){
8              EulerTour(u, node);
9          }
10     }
11     path.pb(node);
12 }
13
14 const int N = 1e5 + 10;
15 int tin[N];
16 int tout[N];
17 int flat[N];
18 int timer = 1;
19 void EulerTour(int node, int parent){
20     tin[node] = timer;
21     flat[timer] = node;
22     timer++;
23     for(int u : G[node]){
24         if(u != parent){
25             EulerTour(u, node);
26         }
27     }
28     tout[node] = timer;
29 }

```

9.2. Binary Lifting

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int up[N][LOG];
4  vector<int> G[N];
5  //Find parents
6  void dfs(int u){
7      for(int v : G[u]){
8          up[v][0] = u;
9          for(int j = 1; j < LOG; j++){
10             up[v][j] = up[up[v][j-1]][j-1];
11         }
12         dfs(v);
13     }
14 }
15 int get_K_parent(int a, int k){
16     for(int j = LOG-1; j >= 0; j++){
17         if(k && (1<<j)){
18             a = up[a][j];
19         }
20     }
21     return a;
22 }

```

9.3. Centroid Decomposition

```

1  //EJERCICIO IMPLEMENTACION XENIA AND TREE
2  //Nodo rojo mas cercano a un nodo v
3
4  const int N = 2e5+10;
5  vector<int>G[N];
6  int tag[N]; // time of discovery of centriodi
7  int fat[N]; // father in centroid decomposition
8  int szt[N]; // size of subtree
9
10 int best[N];
11 int niv[N];
12 int dist[30][N]; //dist[nivel][nodo];
13
14 void dfs_level(int node, int p, int niv, int prof){
15     dist[niv][node] = prof;
16     for(auto y : G[node]){
17         if(y != p and tag[y] < 0){
18             dfs_level(y, node, niv, prof+1);
19         }
20     }
21 }
22 // -----CENTROID-----
23 int calcsz(int node, int p){
24     szt[node] = 1;
25     for(auto y : G[node]){
26         if(y != p and tag[y] < 0){
27             szt[node] += calcsz(y, node);
28         }
29     }
30     return szt[node];
31 }
32 int ccnt = 0;
33 void cdfs(int node, int p, int nivel = 0, int sz = -1){
34     if(sz < 0) sz = calcsz(node, -1);
35     for(auto y : G[node]){
36         if(tag[y] < 0 and szt[y] * 2 >= sz){
37             szt[node] = 0;
38             cdfs(y, p, nivel, sz);
39             return;
40         }
41     }
42     tag[node] = ccnt++;
43     fat[node] = p;
44     niv[node] = nivel;
45     dfs_level(node, -1, nivel, 0);
46     for(auto y : G[node]){
47         if(tag[y] < 0){
48             cdfs(y, node, nivel+1);
49         }
50     }
51 }
52
53 int n;
54 void centroid(){
55     ccnt = 0;
56     repl(1,1,n+1){
57         tag[i] = -1;
58     }
59     cdfs(1, -1);
60 }
61
62 //-----
63
64 void update(int node, int rojo){
65     int nivel = niv[node];
66     int cost = dist[nivel][rojo];
67     best[node] = min(best[node], cost);
68     int p = fat[node];
69     if(p > 0){
70         update(p, rojo);
71     }
72 }
73
74 int improve(int node, int cur){
75     if(node <= 0){
76         return 1e9;
77     }
78     int nivel = niv[node];
79     int cost = best[node] + dist[nivel][cur];
80     int p = fat[node];
81     return min(cost, improve(p, cur));
82 }
83
84 void solve(){

```

```

85     int qq;
86     cin >> n >> qq;
87     repl(i,0,30){
88         repl(j,i,n+1){
89             dist[i][j] = -1;
90             best[j] = 1e9;
91         }
92     }
93     repl(i,0,n-1){
94         int a, b;
95         cin >> a >> b;
96         G[a].push_back(b);
97         G[b].push_back(a);
98     }
99     centroid();
100    update(1, 1);
101    while(qq--){
102        int q, node;
103        cin >> q >> node;
104        if(q == 1){
105            update(node, node);
106        }
107        else{
108            int res = best[node];
109            int res2 = improve(node, node);
110            cout << min(res, res2) << '\n';
111        }
112    }
113 }
114
115 int main(){
116     velocito;
117     int t = 1;
118     // cin >> t;
119     while(t--){
120         solve();
121     }
122     return 0;
123 }
124 //Sabrossus
125 //TC Argentina

```

9.4. Lowest Common Ancestor

```

1  const int N = 1e5;
2  const int LOG = 20;
3  vector<int> G[N];
4  int depth[N];
5  int up[N][LOG];
6  // Assign levels and get parents
7  void dfs(int u){
8      for(int v : G[u]){
9          depth[v] = depth[u] + 1;
10         up[v][0] = u;
11         for(int j = 1; j < LOG; j++){
12             up[v][j] = up[up[v][j-1]][j-1];
13         }
14         dfs(v);
15     }
16 }
17 int get_lca(int a, int b){
18     if(depth[a] < depth[b]) swap(a,b);
19     int k = depth[a] - depth[b];
20     for(int j = LOG-1; j >= 0; j--){
21         if(k & (1<<j)){
22             a = up[a][j];
23         }
24     }
25     if(a == b) return a;
26     for(int j = LOG-1; j >= 0; j--){
27         if(up[a][j] != up[b][j]){
28             a = up[a][j];
29             b = up[b][j];
30         }
31     }
32     return up[a][0];
33 }

```

10. Utilities

10.1. Int128

```

1  // --- LECTURA B SICA ---
2  __int128 read128() {
3      string s;
4      cin >> s;
5      __int128 n = 0;
6      int sign = 1;
7      size_t start = 0;
8
9      if (!s.empty() && s[0] == '-') {
10         sign = -1;
11         start = 1;
12     }
13
14     for (size_t i = start; i < s.length(); ++i) {
15         n = n * 10 + (s[i] - '0');
16     }
17
18     return n * sign;
19 }
20
21 // --- IMPRESION B SICA ---
22 void print128(__int128 n) {
23     if (n == 0) {
24         cout << 0;
25         return;
26     }
27     if (n < 0) {
28         cout << '-';
29         n = -n;
30     }
31     string s;
32     while (n > 0) {
33         s += (char)('0' + (n % 10));
34         n /= 10;
35     }
36     reverse(s.begin(), s.end());
37     cout << s;
38 }

```

10.2. Merge Sort

```

1  // Rango de uso: 0-indexed.
2  // Llamar en el solve como: mergeSort(v, 0, n - 1);
3  ll ans = 0;
4
5  void mergeSort(vector<ll> &v, int low, int high) {
6      if (low == high) return;
7
8      int mid = (low + high) >> 1;
9      mergeSort(v, low, mid);
10     mergeSort(v, mid + 1, high);
11
12     queue<ll> L, H;
13     for (int i = low; i < mid + 1; i++) {
14         L.push(v[i]);
15     }
16     for (int i = mid + 1; i < high + 1; i++) {
17         H.push(v[i]);
18     }
19
20     for (int i = low; i < high + 1; i++) {
21         if (L.size() == 0) {
22             v[i] = H.front();
23             H.pop();
24         }
25         else if (H.size() == 0) {
26             v[i] = L.front();
27             L.pop();
28         }
29         else {
30             if (L.front() <= H.front()) {
31                 v[i] = L.front();
32                 L.pop();
33             }
34             else {
35                 v[i] = H.front();
36                 H.pop();
37                 ans += L.size(); // Inversion detectada
38             }
39         }
40     }
41 }

```

```

40     }
41 }

```

10.3. Prefix Sum 2D

```

1  //1 indexed
2  const int N = 1005;
3  int v[N][N];
4  int acu[N][N];
5  int n, m;
6
7  void solve() {
8      //limpiar la acu para cada TC
9
10     // Construcción: acumula 1 si el elemento es exactamente 1
11     for (int i = 1; i <= n; i++) {
12         for (int j = 1; j <= m; j++) {
13             int val = (v[i][j] == 1 ? 1 : 0);
14             acu[i][j] = val + acu[i - 1][j] + acu[i][j - 1] - acu[i - 1][j - 1];
15         }
16     }
17
18     // Esquinas de consulta:
19     // (a, b) -> Superior Izquierda
20     // (c, d) -> Inferior Derecha
21     int a = 2, b = 2, c = 4, d = 4; // Valores de ejemplo
22
23     int ans = acu[c][d] - acu[a - 1][d] - acu[c][b - 1] + acu[a - 1][b - 1];
24     cout << ans << '\n';
25 }

```

10.4. Random

```

1  mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
2  // mt19937_64 para 64 bits usar uint64_t o unsigned long long
3  cout << rng() << endl;
4  shuffle(v.begin(), v.end(), rng);

```

10.5. Plantilla Python

```

1  import sys
2  from bisect import bisect_left, bisect_right, insort
3  from collections import defaultdict, Counter, deque
4  from heapq import heappush, heappop, heapify
5  from math import gcd, lcm, isqrt
6
7  input = sys.stdin.readline
8
9
10 def solve():
11     n = int(input()) #leer numero
12     a,b,c= map(int, input().split()) #leer numeros
13     s = input().strip() #leer cadenas
14     v = list(map(int, input().split())) #leer lista de enteros
15
16
17     t = 1
18     # t = int(input())
19     for _ in range(t):
20         solve()
21
22     # C++                                Python
23
24     # vector                                list
25     # array                                list
26     # string                                str
27
28     # set                                    set
29     # map                                    dict
30     # multiset                              Counter
31     # unordered_map                         dict
32     # unordered_set                         set
33
34     # priority_queue                        heapq
35     # queue                                 deque
36     # deque                                 deque
37

```

```
38 # lower_bound          bisect_left
39 # upper_bound          bisect_right
40 # sort                 list.sort()
41 # binary_search        bisect_left/right
42
43 # pair                 tuple
44 # struct               class / dataclass
45
46 # gcd                  math.gcd
47 # lcm                  math.lcm
48
49 # push_back            append
50 # push_front           appendleft
51 # pop_back             pop
52 # pop_front           popleft
53
54 # size()               len()
55 # empty()              not x
```

10.6. Pragma

```
1 #pragma GCC optimize("O3,unroll-loops")
2 #pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")
```